WASP-50b

R-0850

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-50_181104` · created 2026-07-28T20:43:31+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458426.7327 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.06 +/- 0.049
Transit depth (Rp/Rs)^2	0.36 +/- 0.59 [%]
Orbital Inclination (inc)	80.75 +/- 2.86
Airmass coefficient 1 (a1)	1.207 +/- 0.067
Airmass coefficient 2 (a2)	-0.114 +/- 0.046
Scatter in the residuals of the lightcurve fit is	5.52 %
Best Comparison Star	#3 - [297, 187]
Optimal Aperture	7.18
Optimal Annulus	13.76
Transit Duration (day)	0.019 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.06 ± 0.049 vs 0.1404 (deviation 1.64σ)
Duration fit vs published	0.019 d vs 0.0754 d (deviation 2.01σ)
Mid-time O–C	-1.8 min
Depth SNR	1.2
Residual scatter	5.52 %

Light curve

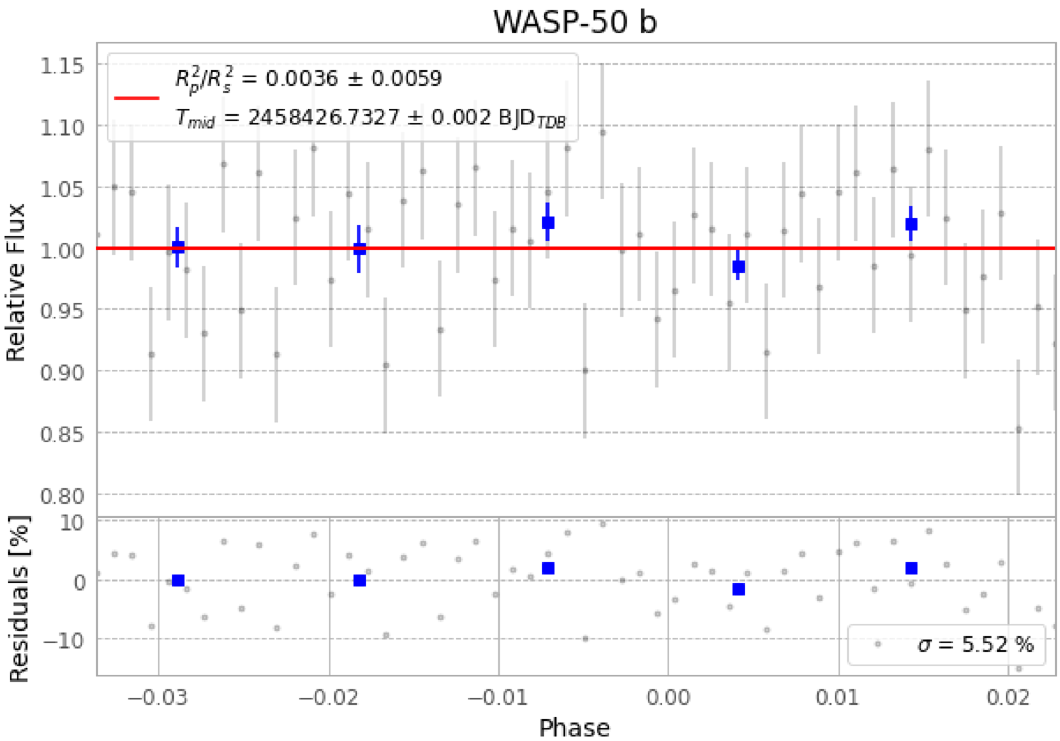

Figure 1 — FinalLightCurve_WASP-50 b_2018-11-03.png (SHA-256 b2ec2581a1dc4354…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [303, 403], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 6b011193bd89c5c5…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

76 FITS frames (50 MB), `WASP-50181104035112.FITS` → `WASP-50181104073612.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-181104.log (1ec88abfe90a…), FinalLightCurve_WASP-50 b_2018-11-03.png (b2ec2581a1dc…), FinalLightCurve_WASP-50 b_2018-11-03.csv (1a8b9e6b1654…)

Compute resources

Wall time 160.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 20:43Z.